Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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This document reports, in full, four robustness checks summarized in the main text, the variants of the candidate measure (Section S5), and further analyses summarized in one sentence in the main text (Section S6). All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption, and Table S7 lists the script behind every result reported in the main text.

Table S1: Panel-size recoverability under alternative correlation bars

The main text calls a panel size "recovering" the full-panel ranking once mean Spearman r against the full panel exceeds 0.90, a bar borrowed from the "excellent" range of intraclass-correlation guidelines (Results). Table S1 reports p^* (the smallest panel size reaching the bar) at four bars, including the main-text bar of 0.90, for all four datasets.

The ratio remains the slowest to recover on every dataset for every bar tested, and the Gini coefficient the second-slowest on Davis, Klaeger, and Anastassiadis. On Metz it ties the entropy at every bar. No bar in $[0.80, 0.95]$ reverses the order of two definitions on any dataset. The bar decides only whether some of them tie.

Table S1: Panel-size p^* (kinases) at four correlation bars. (`pstar_bar_sensitivity.py`)

Davis (433 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	80	50	350
bar = 0.85	50	80	50	380
bar = 0.90	50	110	50	410
bar = 0.95	50	170	80	410

Klaeger (343 targets)	entropy	gini	s_score	ratio
bar = 0.80	50	260	50	320
bar = 0.85	50	260	80	320
bar = 0.90	80	290	110	343
bar = 0.95	140	320	200	343

Anastassiadis (300 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	170	50	260
bar = 0.85	50	170	50	260
bar = 0.90	50	200	50	290
bar = 0.95	80	260	80	290

Metz (172 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets to check whether they behave as additional members of the ratio family (WS, RS) or the distribution-based family (CATDS), as claimed in the Discussion. (`bosc_catds_comparison.py`)

Table S2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.804	0.878	0.509	0.955 / 0.960
Klaeger	222	0.909	0.892	0.681	0.824 / 0.985
Anastassiadis	178	0.959	0.891	0.696	0.945 / 0.965
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.80$–0.96), supporting their classification alongside $R(k)$. The CATDS proxy instead correlates more strongly with entropy and Gini than with the ratio on every dataset, supporting the statement in Methods that CATDS behaves like a member of the distribution-based family.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

Table S3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.417	0.952	0.456
entropy(literal)	0.417	1.000	0.424	0.336
Gini	0.952	0.424	1.000	0.343
ratio	0.456	0.336	0.343	1.000

Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.570	0.722	0.223
entropy(literal)	0.570	1.000	0.688	0.789
Gini	0.722	0.688	1.000	0.616
ratio	0.223	0.789	0.616	1.000

Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.291	0.821	0.193
entropy(literal)	0.291	1.000	0.385	0.758
Gini	0.821	0.385	1.000	0.155
ratio	0.193	0.758	0.155	1.000

Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.881	0.345	0.489
entropy	0.881	1.000	0.631	0.526
Gini(native)	0.345	0.631	1.000	0.314
ratio	0.489	0.526	0.314	1.000

On Klaeger and Metz, literal entropy correlates more strongly with the ratio than with either S-score or Gini – under this operationalization, entropy would not be classified in the distribution-based family at all on either of these two datasets. On Davis, literal entropy decouples from both families (consistent with Figure 1 of the main text). The native-Gini check on Anastassiadis

preserves the pattern only partly: the native Gini correlates with the entropy (0.63) but weakly and about equally with the S-score (0.35) and the ratio (0.31).

Table S4: Candidate measure cross-dataset validation

Table S4 reports three checks for each dataset. Orientation is the correlation with the number of active kinases, where more negative means more correctly oriented toward "fewer active targets = more selective". D3 is reported as in Methods, as the percentage of pairs of scored compounds whose order reverses somewhere in $B = [x_0, \tau^*]$. It is given for the emphasis parameter β with the gate x_0 fixed (the featured measure), for the gate x_0 itself swept as a free parameter (the like-for-like comparison with the entropy's baseline, Figure 6 of the main text), and for the entropy's own baseline. D4 is the percentage of compounds for which the score does not rise when one kinase is added at the activity threshold τ^*, the strongest binding D4 allows and so the addition most likely to raise the score. B is $[5.0, 6.0]$ on Davis and Klaeger, $[4.0, 6.0]$ on Metz, and $[20, 50]\%$ inhibition on Anastassiadis, and the smoothing width is $T = \tau^* - x_0$ on every dataset (Methods). D3 passes at $\varepsilon = 1\%$. (`candidate_validation.py`, `d3_pair_reversals.py`)

Table S4: Candidate measure: orientation, D3 (pairs reversing in B, %), and D4, across all four datasets, over the n compounds that D1 allows to be scored.

Dataset	n	orientation r	D3, gate fixed	D3, gate swept	D3, entropy	D4 (%)
Davis	68	−0.969	0.48	8.52	8.82	100.0
Klaeger	206	−0.951	0.54	10.57	11.20	100.0
Anastassiadis	134	−0.936	0.43	12.49	10.27	100.0
Metz	699	−0.783	0.72	22.45	22.60	100.0

The "gate swept" and "entropy" columns confirm, on every dataset, that the candidate's D3 satisfaction, like the entropy's, rests on fixing its activity boundary at x_0. Swept over the same interval, the candidate's gate reverses about as many pairs as the entropy's baseline does, and the emphasis parameter, with the gate fixed, reverses fewer than one pair in a hundred. Removing the gate inverts the measure's orientation entirely (gated $r = -0.951$ vs. ungated $r = +0.93$ against n_{active} on Klaeger), which is why the gate is required for correct orientation.

S5: Variants of the candidate measure

The featured candidate is one member of a family of measures. Three axes of variation are reported here: the floor, the Rényi order, and the smoothing width T. Each variant is checked against D1–D4 as in the main text. D4 is proved only for the featured measure (Shannon order, $\beta = x_0$, $T \geq 0.58\,(\tau^* - x_0)$), and is otherwise checked by adding one kinase at τ^* to every compound. (`family_construction.py`, `family_panelsize.py`, `candidate_validation.py`)

Quantile floor. Replacing the fixed floor x_0 with a per-compound low quantile of the profile (the 5th to 15th percentile) gives a variant that is invariant to shifts of the readout, but on these data it is not a separate test: on Klaeger every such percentile of every compound's profile equals the floor, and on Davis it does for all but at most two compounds, so the variant ranks compounds exactly as the featured measure does (Spearman 1.000).

Rényi order. Generalizing the Shannon functional to Rényi order $\alpha > 0$,

$$-H_\alpha(p) = \frac{1}{\alpha - 1} \log_2 \sum_i p_i^\alpha,$$

gives a family that recovers the featured measure at $\alpha = 1$. D1 holds for every order, since the gate is applied before the entropy is computed, and D2 holds vacuously. D4 holds on every compound with one kinase added at τ^* for each order tested. D3 separates the orders (Table S5): the orders 0.5 and 1 pass on both datasets, $\alpha = 2$ is borderline (it passes on Davis and narrowly fails on Klaeger), and $\alpha = 3$ fails on both.

Table S5: Rényi-order variants on Davis and Klaeger: D3 (pairs reversing in B, %), agreement with the Shannon ranking (Spearman), and panel size p^* on Klaeger (smallest panel reaching mean $r > 0.90$). On Metz every order also reaches $p^* = 80$. (`family_construction.py`, `family_panelsize.py`)

Order	D3, Davis	D3, Klaeger	Spearman vs. $\alpha = 1$	p^*, Klaeger
$\alpha = 0.5$	0.61	0.13	0.999–1.000	80
$\alpha = 1$	0.48	0.54	—	80
$\alpha = 2$	0.92	1.09	0.999	80
$\alpha = 3$	1.80	1.68	0.997–0.998	80

The order does not affect panel recoverability. Every order tested, from 0.01 to 3, reaches $p^* = 80$ on both Klaeger and Metz, and the mean convergence curves differ by at most 0.003 between orders at every panel size. We feature $\alpha = 1$ because it is the Shannon entropy, for which D4 is proved.

Smoothing width. The featured measure uses $T = \tau^* - x_0$. On Klaeger, over $T = 0.5$, 0.75, 1, 1.5, and 2 (in units of $\tau^* - x_0 = 1$), orientation is unchanged ($r \approx -0.95$) and the rankings agree closely across all five values (worst-case pairwise Spearman 0.999). D3 passes for $T \geq 0.75$ (pairs reversing as the emphasis baseline is swept: 0.93%, 0.54%, 0.17%, 0.04% for $T = 0.75$, 1, 1.5, 2) and narrowly fails at $T = 0.5$ (1.94%), which also falls below the D4 bound $T \geq 0.58$. As $T \to 0$ the softplus converges to the hard cutoff, recovering the entropy together with its D3 and D4 failures.

S6: Additional analyses

The analyses below are summarized in one sentence in the main text, and are given here in the order the main text cites them.

Klaeger restricted to protein kinases. The 343 Klaeger targets are the proteins the Kinobeads pulldown quantified, and 120 of them are not protein kinases: 46 cyclins and other subunits or adaptors of kinase complexes (for example CCNA2–CCNT2, CKS1B, the AMPK and PKA regulatory subunits, and the AP2 adaptor subunits), 10 small-molecule and lipid kinases, 20 other enzymes (for example COMT, NQO2, TOP2B, PYGB, PYGL and PTPN1), and 44 other proteins, including ribosomal proteins and SLC transporters. Kinobeads co-capture complex partners with their kinases, so a CDK hit can also count its cyclin. These columns hold 519 of the 2,763 active entries (18.8%), in 117 of the 222 compounds, and five compounds (AZD-2014, AZD-8186, SGX-523, TG-100115, volitinib) are active only on them. Table S6 recomputes the main Klaeger results on the 223 protein kinases. Among compounds with an active kinase, the two-family structure, the D3 verdicts and all but one of the instability relations are essentially unchanged, and the candidate and the entropy still rank compounds almost identically and reach the same p^*. Three results move. With all compounds included, the five new zero-active compounds pull the Gini toward the ratio (0.655 against 0.515 with the S-score), so, as in the main text, its placement on Klaeger depends on the zero-active compounds. The skewness–Gini association weakens from 0.35 to 0.24 (still $p < 0.001$). The S-score and Gini reach p^* at fewer kinases (80 and 200), partly because the full panel is smaller, and the ratio still needs the full panel.

Table S6: Klaeger results on the full 343-target panel and on its 223 protein kinases. Correlations are Spearman r; "active" rows re-rank the compounds with an active kinase among themselves. (`klaeger_protein_kinases.py`)

Result	All 343 targets	223 protein kinases
Zero-active compounds	16	21
Active entries (% of matrix)	3.6	4.5
Table 1, all compounds: S–E, S–G, E–G	0.910, 0.722, 0.840	0.897, 0.515, 0.661
Table 1, all compounds: R against S, E, G	0.223, 0.455, 0.616	0.092, 0.349, 0.655
Table 1, active: S–E, S–G, E–G	0.913, 0.911, 0.998	0.898, 0.896, 0.998
Table 1, active: R against S, E, G	0.418, 0.621, 0.634	0.381, 0.616, 0.629
$r(\mathrm{S},\mathrm{G}) - r(\mathrm{G},\mathrm{R})$, active, 95% interval	$[0.19, 0.37]$	$[0.17, 0.37]$
Pooled rank σ, zero-active vs. active	74.0 vs. 31.2	78.8 vs. 32.0
Table 2: n_{active}, entropy instability	-0.381	-0.369
Table 2: top_1–top_2 gap, ratio instability	-0.341	-0.353
Table 2: active range, S-score instability	-0.354	-0.352
Table 2: skewness, Gini instability	$+0.346$	$+0.241$
D3, pairs reversing (%): S-score, entropy, Gini	8.3, 11.2, 11.5	8.5, 12.1, 12.4
D3, candidate: emphasis baseline, gate (%)	0.54, 10.6	0.40, 11.4
Candidate vs. entropy, scored compounds	0.998	0.999
p^*: candidate, entropy, S-score, Gini, ratio	80, 80, 110, 290, 343	80, 80, 80, 200, 223

Klaeger confidence filter. The high-confidence filter removes 793 of the 5,916 drug–target rows in the table (13.4%), one of which has no numeric K_d^{app}. A removed pair is treated identically to one with no detected binding and assigned the same $\mathrm{p}K_d = 5.0$ background value. Of the 792 low-confidence pairs with a numeric K_d^{app}, 561 fall within the 222×343 high-confidence matrix. The

other 231 involve 7 compounds and 177 kinases with no high-confidence measurement. Adding the 561 pairs to the same matrix changes only 0.74% of its entries, shifts every cell of the Klaeger triangle of main-text Table 1 by at most 0.05, and leaves every Klaeger panel-size p^* value unchanged. Retaining all 792 pairs also adds those compounds and kinases, which changes the panel itself. It shifts the Klaeger cells of Table 1 by up to 0.12 (S-score against ratio, from 0.22 to 0.34) and leaves the two-family structure intact. (`klaeger_filter_sensitivity.py`)

Skewness and Gini instability. Among the 161 Klaeger compounds with at least three active kinases, skewness correlates with the share of a compound's weight carried by kinases that the β sweep crosses ($r = 0.45$), and that share predicts Gini instability with the number of active kinases held fixed (partial $r = 0.71$). Holding it fixed as well lowers the skewness association from 0.57 to 0.31, so this mechanism accounts for part of the association. (`claim_checks.py`, section 16)

Type 3 compounds. Five of the ten largest entropy–ratio disagreements among active Klaeger compounds have a top_1–top_2 gap above 1, for example dovitinib ($n_{\text{active}} = 26$, gap = 1.935) and BMS-690514 ($n_{\text{active}} = 20$, gap = 1.985). The ratio ranks these five 22nd to 60th of 222, and the entropy and Gini rank them 157th to 191st. (`claim_checks.py`, section 5; `klaeger_selectivity_results.csv`)

Most stable compounds. On Klaeger the most stable compounds are among the most promiscuous. XL-228 ($\sigma = 7.0$) and PF-03814735 ($\sigma = 7.8$), with 55–57 active kinases, have median ranks of 208–220 of 222 under every definition, and mean instability falls from about 31–33 for compounds with 1 to 40 active kinases to 16 above 40. On Davis, compounds with few active kinases are the most stable on average (mean $\sigma = 3.1$ for one or two, 4.4 for three to five). The most stable, neflamapimod (main-text Figure 4, left, $\sigma = 1.8$), has two, with median ranks of 3–5.5 under the four definitions. The 57 Davis compounds with more than 20 active kinases are less stable (mean $\sigma = 9.1$, against 5.3 for the other 11). The S-score, entropy, and Gini give the five broadest median ranks of 63–68 of 68, and the ratio gives them 29–52. Without the ratio's configurations, the mean instability of the 57 broad compounds falls from 9.1 to 5.1. (`claim_checks.py`, sections 7 and 18)

Table S7: Provenance of reported results

Every result in the main text is printed by a script in the code repository. Table S7 gives, for each result, the script that computes it. Each script saves its printed output as `outputs/<script>.txt`; numbers in parentheses are numbered sections of that output, and a few results are read from the CSV file the script writes, named in parentheses. Random steps are seeded, so rerunning a script reproduces its output.

Table S7: Provenance of reported results, grouped by manuscript section.

Location	Result	Script (output section)
Methods, Datasets; Candidate	Active fractions; stricter S-score cutoff on Davis; missing entries and noise floor on Anastassiadis; Metz coverage per compound	`dataset_descriptive_stats.py`
Methods, Datasets	Active fractions of the 46 compounds shared by Davis and Klaeger	`claim_checks.py` (9)
Methods, Datasets; Section S6	Klaeger restricted to protein kinases (Table S6)	`klaeger_protein_kinases.py`
Methods, Datasets; Section S6	Klaeger confidence filter: rows removed, relaxed-filter sensitivity	`klaeger_filter_sensitivity.py`
Methods, Datasets	Metz coverage distribution	`metz_coverage.py`
Methods, Datasets	Metz rankings vs number of measured kinases	`claim_checks.py` (10)
Methods, Definitions	Compounds with no kinase above the threshold; ratio off-target cap	`dataset_descriptive_stats.py`
Methods, Ratio; Discussion	Bosc et al. scores and CATDS proxy (Table S2)	`bosc_catds_comparison.py`
Methods, Faithfulness; Figure 1	Published-form entropy and Gini (Table S3); operationalization scatter	`original_operationalization_robustness.py`, `metric_fidelity_robustness.py`
Methods, Rank Stability	Table 1 with mean instead of median ranks	`claim_checks.py` (3)
Methods, Candidate	Gate and smoothing-width bound for D4, including every β in B	`d4_entropy_vs_candidate.py` (1)
Table 1	Correlations between definitions (Davis, Klaeger)	`selectivity_analysis.py`, `klaeger_analysis.py`
Results, Two Families	Klaeger correlations without zero-active compounds; Gini bootstrap	`claim_checks.py` (2)
Figure 2; Results, Replication	Correlations on four datasets; replicated instability relations	`additional_datasets_analysis.py` (`cross_dataset_summary.csv`)
Results, Replication	Within-family and ratio ranges, all and active-only compounds	`claim_checks.py` (11)
Results, Replication; D1	Zero-active contrast within each definition's sweep	`claim_checks.py` (13)
Results, Replication; Type 1; D1	Pooled zero-active contrast, with and without undefined configurations; where each definition ranks zero-active compounds	`claim_checks.py` (20)
Table 2; Figure 3	Profile features vs within-family instability	`instability_by_family.py`
Results, Instability	Overall rank instability, mean and range	`klaeger_analysis.py`, `selectivity_analysis.py`

Location	Result	Script (output section)
Results, Instability	Entropy instability vs share of kinases crossing the baseline sweep	`claim_checks.py` (8)
Results, Instability	Range vs instability holding the number of active kinases fixed	`claim_checks.py` (15)
Results, Instability; Section S6	Skewness association and threshold weight	`claim_checks.py` (16)
Results, Instability	Range vs S-score instability on the 46 shared compounds	`claim_checks.py` (17)
Results, Instability	Table 2 with a Bonferroni correction	`claim_checks.py` (21)
Results, Instability	Anastassiadis gap–ratio relation without saturated compounds	`claim_checks.py` (4)
Results, Instability	Undefined skewness and range: placeholder sensitivity	`feature_placeholder_sensitivity.py`
Results, Instability	Further candidate features	`instability_predictor_candidates.py`
Results, Sources	The 20 most unstable Klaeger compounds by type	`claim_checks.py` (6)
Results, Type 1; Section S6	Instability and median ranks of individual Klaeger compounds	`klaeger_analysis.py` (`klaeger_selectivity_results.csv`)
Results, Type 1	Were the zero-active compounds' targets quantified	`claim_checks.py` (1)
Results, Type 3; Section S6	Ten largest entropy–ratio disagreements	`claim_checks.py` (5)
Section S6	Instability by number of active kinases; most stable compounds	`claim_checks.py` (7, 18)
Figure 4	Binding profiles of three Davis compounds	`binding_profiles.py`
Figure 5; Table S1	Panel-size recovery and p^* at four bars	`panel_size_analysis.py`, `pstar_bar_sensitivity.py`
D1; Figure 6	Zero-active instability under each convention; D1 and D3 panels	`undefined_score_convention.py`, `failure_mode_illustration.py`
Table 3; D2	Ratio at near-tied top targets	`d2_near_tie_sensitivity.py`
Table 3; D3	Pairs reversing in B	`d3_pair_reversals.py`
Candidate; Table S4	Orientation, D3, D4 at τ^*, smoothing-width sweep	`candidate_validation.py`
Candidate	Dependence on the activity threshold τ^*	`claim_checks.py` (22)
Candidate; Figure 7	Agreement with the entropy; panel-size recovery, gated and ungated	`candidate_measure.py`
Candidate	Gini p^* without compounds whose Gini is undefined	`claim_checks.py` (14)
Candidate; Section S5	Variants of the candidate and their panel sizes; quantile floor	`family_construction.py`, `family_panelsize.py`
Smooth Hinge	Entropy and candidate under one added kinase at τ^*; worked example	`d4_entropy_vs_candidate.py` (2–6)
Smooth Hinge	Realistic panel expansion	`claim_checks.py` (19)
Discussion	Platform vs definition on the 46 shared compounds	`davis_klaeger_overlap.py`, `platform_vs_definition.py`
Discussion	Instability vs cross-platform rank change	`claim_checks.py` (9)

Location	Result	Script (output section)
Discussion	Maximum affinities of the zero-active compounds	`failure_mode_illustration.py`
Discussion	Compounds shared by Klaeger and Anastassiadis	`claim_checks.py` (12)
Discussion, Limitations	Clinical outcomes	`clinical_outcomes_correlation.py`, `faers_pull.py`